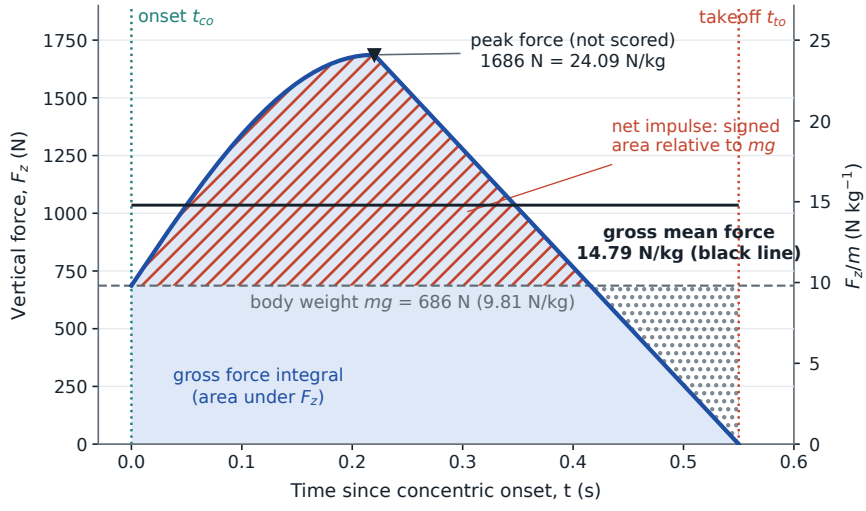
**a** **Concentric-phase force–time curve (analytical, not athlete data)**




**b** **Three distinct quantities**

**Gross mean concentric force**

$$Y_F = \frac{1}{mT}\int_{t_{co}}^{t_{to}} F_z\,dt$$

Basis of the scored force target

**14.79** N/kg

**Peak force**

$$\max_t F_z/m$$

Not the scored target

**24.09** N/kg

**Gravity-adjusted net impulse**

$$Y_J = \frac{1}{m}\int_{t_{co}}^{t_{to}} (F_z - mg)\,dt$$

Separate scored outcome

**2.74** m/s

Illustrative: m = 70 kg, g = 9.80665 m s$^{-2}$, A = 1000 N, t_p = 0.22 s, T = 0.55 s.
t ≤ t_p: F = mg + A sin(πt / 2t_p)
t > t_p: F = (mg + A)[1 − (t − t_p)/(T − t_p)]

**c** **Assessment chronology relative to the index exposure (native contract and public synthetic task are separate)**

Assessment = mean of 3 valid trials · baseline = mean of the 2 latest valid assessments · no athlete data plotted

